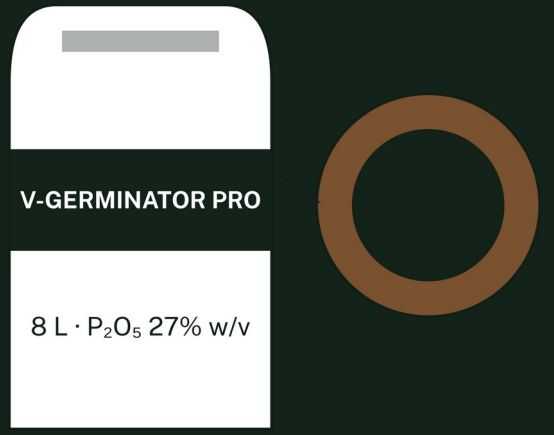
DEALER TRAINING · NOVEMBER 2026

V-Germinator Pro



A liquid phosphate with 27 g of P_2O_5 in every 100 ml, applied in the irrigation water at the stages VAN gives for each crop group.

VAN wrote this deck for its dealers and field team. A dealer can repeat its words to a grower.

THE 30-SECOND EXPLANATION

What a dealer says to a grower in 30 seconds.

V-Germinator Pro is a liquid phosphate in an 8 L pack, with 27 g of P_2O_5 in every 100 ml. It also contains small amounts of nitrogen, potash, magnesium, sulfur and iron. It is black, acidic (pH 2 to 4) and fully soluble, in a plant-based carrier. On calcareous soil, phosphate from a granule meets calcium within centimetres and can be locked before the root reaches it. A liquid goes with the irrigation water to the root at the stage the crop needs it. VAN's guidance gives the stages by crop group: fruiting vegetables 1 week after planting, then before and after flowering; potato at tuber initiation; maize at 2 to 3 leaves; orchards about 40 days after the first phosphate application. It goes in the irrigation water, on top of the basal phosphate, never instead of it, and never with zinc.

The 4 points it must cover

1. The pack: 8 L, P_2O_5 27% weight per volume (w/v) with N 4.5, K_2O 2, Mg 0.5, S 2, Fe 0.5. PS 5469-2020. 2.16 kg of P_2O_5 in a pack.
2. The soil: calcareous soil locks a granule's phosphate before the root arrives. A liquid reaches the root with the water, at the stage the crop needs it.
3. The stages: by crop group, in VAN's guidance, always in the irrigation water. The 40-day booster is for orchards only.
4. The 2 rules: on top of the basal phosphate, never instead of it. Never in the same tank as zinc.

Explain the product in simple words a grower uses, and keep to the figures on these slides. Every figure in this 30-second explanation is on the pack or in the named evidence on the slides that follow.

THE PACK

What the pack says, and what it is registered as.

8 L

the pack on the price sheet. 2.16 kg of P_2O_5 in every pack, from the declared analysis.

P_2O_5 27%

w/v: 27 g in every 100 ml, 270 g in a litre. The headline nutrient.

N 4.5 · K_2O 2

nitrogen 45 g and potash (K_2O) 20 g per litre. Small shares; phosphate is the main nutrient.

Mg 0.5 · S 2 · Fe 0.5

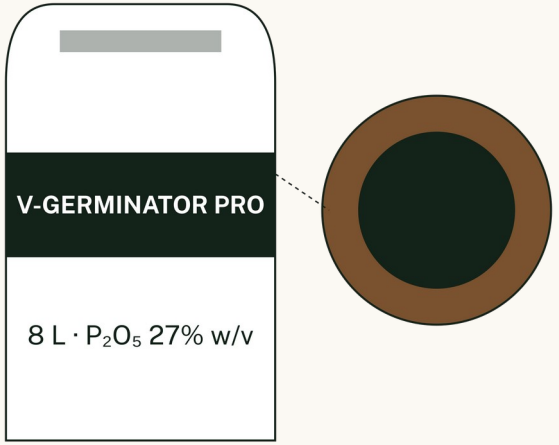
magnesium 5 g, sulfur 20 g and iron 5 g per litre, all declared. Iron is added because high phosphate and lime both reduce the iron available to the plant.

pH 2 to 4

acidic, black, 100% soluble, as VAN's product data gives it. The acid keeps the phosphate in solution until it reaches the root.

Fertigation

the route a dealer gives: through drip or flood irrigation water to the root zone. Drone spraying of V-Germinator Pro is done only through VAN.



Registered as	Pakistan Standard	Manufacturing licence
V-Germinator Pro	PS 5469-2020	CM/L-4300/2026

If a grower asks about registration, give the licence number. Do not give an expiry date for the licence.

A GROWER'S QUESTION

“I put DAP in at sowing. A few weeks after sowing, the crop is slow and the leaves have a purple edge. The DAP is in the ground. Why is the crop not getting it?”

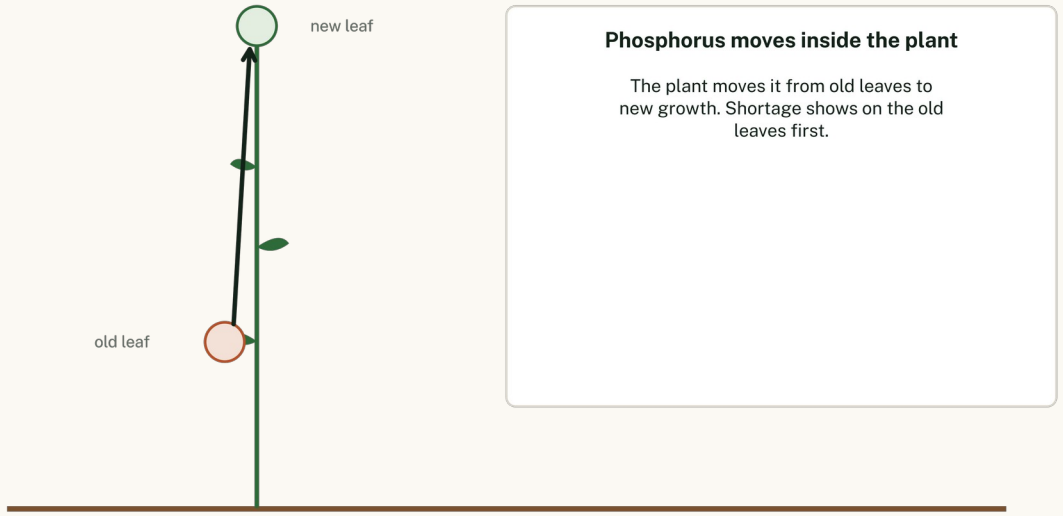
The grower applied the basal phosphate. The problem is where it went: on calcareous soil the phosphate from the granule was locked before the root reached it, and the crop needs phosphorus again. The answer has 3 parts: what calcareous soil does to a granule, the stages a liquid is made for, and what in the pack keeps the phosphate available to the root.

The next 3 slides give the 3 answers, 1 idea on each slide.

ANSWER 1 OF 3

On calcareous soil, phosphate from a granule is locked by calcium before the root arrives.

Phosphate hardly moves in soil. Above pH 8, phosphate dissolving out of a granule is locked by calcium within a few centimetres, before a root arrives. The calcium phosphates that form first are still fairly available; the phosphate becomes unavailable as they slowly change into forms the root cannot use. So a granule has to be exactly where the root will grow, and its phosphate has to stay available for the weeks until the root gets there. In the Punjab soil survey (770,160 samples, tested in about 2016 to 2018), 69.1% of samples had a pH above 8 and 59.1% tested below 7 ppm available phosphorus. A liquid does not sit in the soil for weeks: it moves with the irrigation water to the wet root zone at the stage the crop needs it.



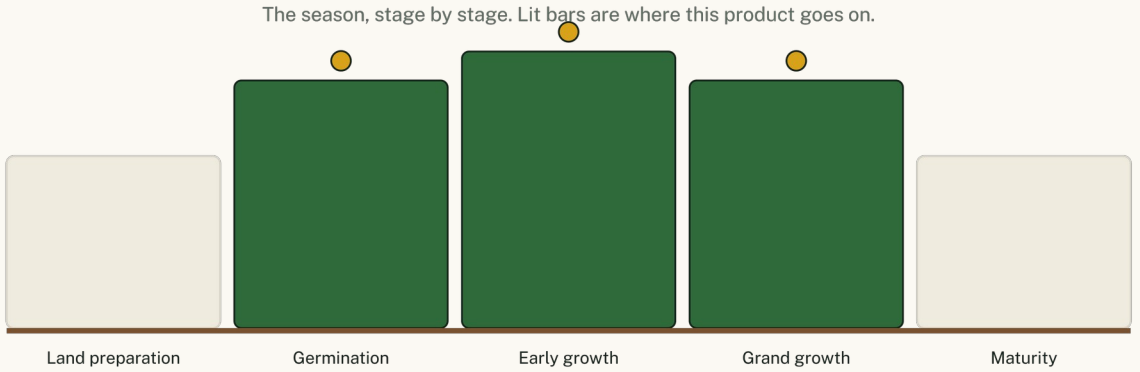
Say it like this: “Your DAP is still in the soil, but it is locked a few centimetres from where the root needed it. This goes with the water to the root.”

The drawing shows phosphorus moving inside the plant, from old leaves to new growth; in the soil it hardly moves. This is general plant and soil chemistry, not a VAN measurement.

ANSWER 2 OF 3

Apply it at the stages VAN's guidance gives for each crop group.

Fruiting vegetables: 1 week after planting, then before and after flowering. Potato: at tuber initiation. Maize: at 2 to 3 leaves. Orchards: about 40 days after the first phosphate application. A young crop needs phosphorus early: phosphorus supplies the energy the plant uses to take up every other nutrient, and it builds the first roots. Growth lost in the first weeks is not made up later. V-Germinator Pro goes on top of the basal phosphate, never instead of it. Applying it at the right stage matters more than the size of the dose.



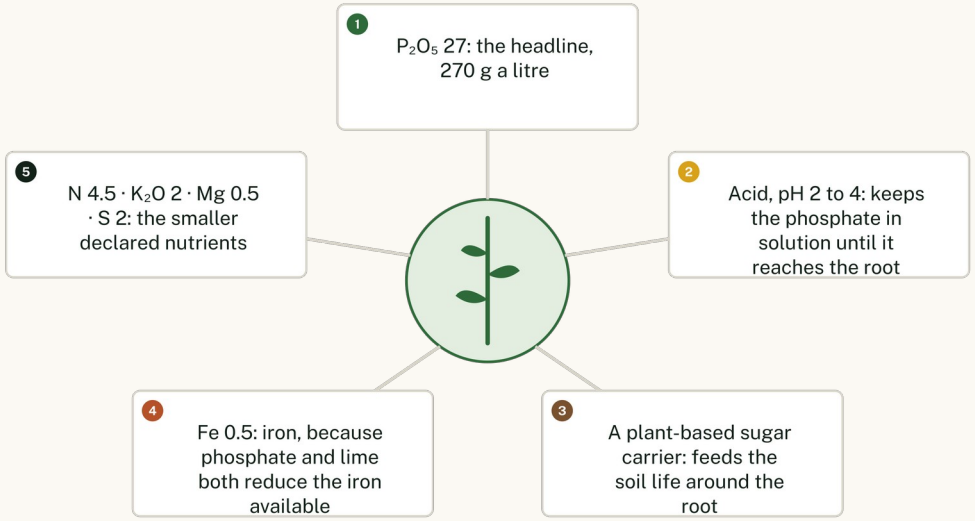
The season bar shows the part of the season the crop-group stages fall in, from soon after planting to flowering. No published crop plan has a row for this product yet.

Say it like this: "Give it at the stage VAN gives for your crop, on top of the basal bag, not in place of it."

ANSWER 3 OF 3

What is in the pack: acid to keep the phosphate in solution, sugar for the soil life, and iron.

The product has a pH of 2 to 4, so its phosphate stays in solution in alkaline irrigation water, in the tank and all the way to the root, instead of reacting with calcium. The carrier is plant-based and rich in sugars. In general, sugars feed the microbes around the root, and microbial activity turns held phosphorus back into forms a root can take up. Iron is declared with it because high phosphate and lime both reduce the iron available to the plant. If the solution is brought towards neutral in the tank, the phosphate drops out. It is also never mixed with zinc, because zinc precipitates with phosphate and can block the line.



The 5 parts of the litre shown in the drawing, from VAN's product data. The drawing shows no percentage of phosphate saved, and VAN claims none.

Say it like this: "The acid keeps the phosphate in solution until it reaches the root. Never put zinc in the same tank."

THE EVIDENCE

What is verified for V-Germinator Pro, and what is not.

No trial of V-Germinator Pro is on VAN's record, and no published crop plan names it. A dealer may quote the analysis on the pack, the licence, the soil survey and the soil chemistry, each only as the right-hand column allows.

Item	What it says	How a dealer may use it
The pack	V-Germinator Pro, P_2O_5 27% w/v (270 g/L), N 4.5%, K_2O 2%, Mg 0.5%, S 2%, Fe 0.5%; black liquid, pH 2 to 4, 100% soluble, plant-based carrier; 8 L.	As fact. Read the 6 nutrients off the pack in hand.
The licence	Pakistan Standard PS 5469-2020 for liquid nutrient supplements; manufacturing licence CM/L-4300/2026.	As fact: quote the number.
VAN's phosphate range	VAN's phosphate range on van.com.pk lists V-Germinator Pro (phosphate booster) as the liquid fertigation product, beside Green Phosphate and Fusion Phosphate (soil, basal) and V-Phosphate (fertigation, solid).	As VAN publishes it: where this product sits in the range.
VAN guidance, 23 July 2026	Crop-group stages: 1 week after planting, then before and after flowering (fruiting vegetables); tuber initiation (potato); 2 to 3 leaves (maize); about 40 days after the first phosphate application (orchards). On top of the basal granule.	Spoken as VAN's guidance. No dose.
The Punjab soil survey (770,160 samples, about 2016 to 2018)	Mean available phosphorus 6.62 ppm; 59.1% of samples below 7 ppm, the low level; 97.9% below 15 ppm, the adequate level; 69.1% above pH 8; mean calcium carbonate 6.53%.	As fact about the soil. Never as product performance.
Not published	A dose per acre. VAN gives the dose for each crop on request.	The dealer asks VAN and does not guess.
Not on file	A V-Germinator Pro trial; a germination, emergence or yield figure; a percentage of phosphate fixed or saved; a crop plan row.	Not said until VAN names a source.

Sources: the pack; the Pakistan Standard and licence register; VAN's phosphate range page on van.com.pk; the VAN farmer price sheet, Aug 2026; the Punjab soil survey (770,160 samples, about 2016 to 2018); VAN guidance, 23 July 2026.

The pack in grams of each nutrient, and its price.

The price is the delivered farmer price from the VAN farmer price sheet, Aug 2026. Nutrients are from the declared analysis, weight per volume: 27% is 270 g in a litre.

V-Germinator Pro	Volume	P_2O_5	N	K_2O	Mg	S	Fe	Rupees
1 litre	1 L	270 g	45 g	20 g	5 g	20 g	5 g	569
1 pack	**8 L**	**2.16 kg**	**360 g**	**160 g**	**40 g**	**160 g**	**40 g**	**4,550**

An 8 L pack carries 2.16 kg of P_2O_5 that goes with the irrigation water to the root, for Rs 4,550. VAN has not published a dose per acre for this product; the dealer asks VAN and does not guess. VAN has no measured figure for the crop's response to it.

V-Germinator Pro 8 L Rs 4,550: the VAN farmer price sheet, Aug 2026; Rs 569 a litre = Rs 4,550 ÷ 8 litres. 270 g = 1,000 ml × 0.27; 2.16 kg = 8 × 0.27; 45 g = 1,000 × 0.045; 5 g = 1,000 × 0.005. Analysis as declared on the pack.

P_2O_5 removed in the harvest, beside what 1 pack carries.

Removal per tonne is from the removal tables of the International Plant Nutrition Institute (IPNI), which the nutrient balance section of van.com.pk uses. The pack is a booster at the crop-group stages; the season's main phosphate is the basal bag applied at land preparation.

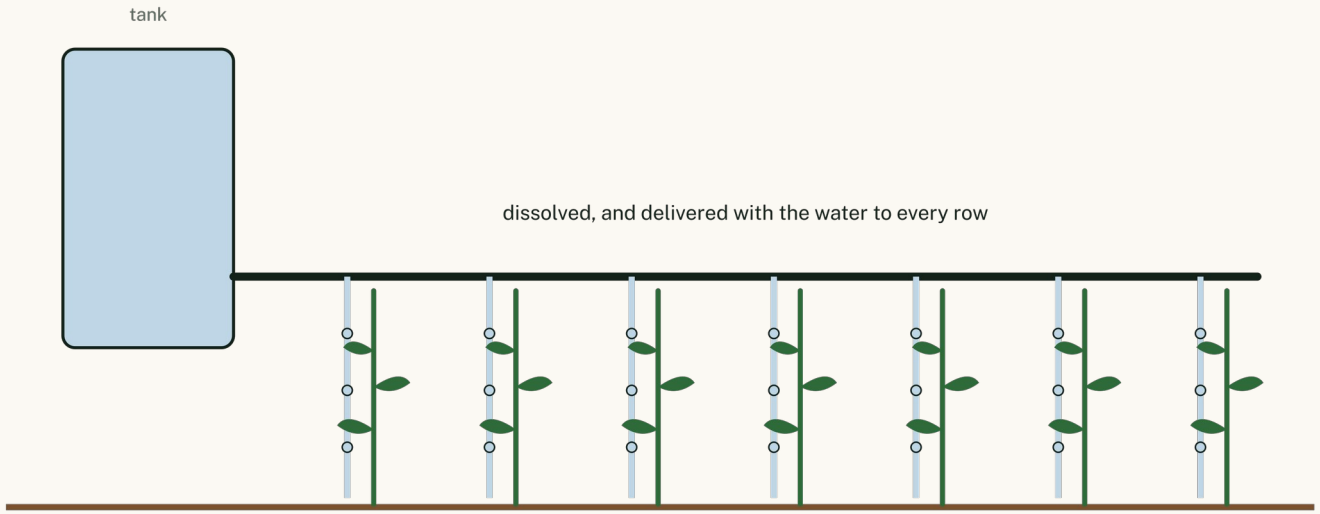
Crop	P_2O_5 removed per tonne	P_2O_5 at a stated yield per acre	What it is set against
Maize grain	6.3 kg	31.5 kg at 5 t per acre (125 maunds)	The basal phosphate at land preparation
Potato tubers	1.5 kg	15 kg at 10 t per acre (250 maunds)	The basal phosphate at land preparation
Wheat grain	8.0 kg	12.8 kg at 1.6 t per acre (40 maunds)	The basal phosphate at land preparation
V-Germinator Pro, 1 pack of 8 L, carries		**2.16 kg**	**A booster at the crop-group stages**

The pack does not supply the crop's phosphate for the season, and VAN does not sell it as if it did. It supplies phosphate in solution at the stages VAN gives for each crop group. Keep the basal bag and add this product at those stages.

Removal per tonne: the IPNI removal tables (metric edition dated 01/15): maize grain 6.3, potato tuber 1.5, wheat grain 8.0 kg P_2O_5 per tonne. The yields in the table are assumed examples per acre, for illustration, not VAN measurements; 1 maund is 40 kg. 2.16 kg = 8 × 0.27.

HOW IT IS APPLIED

Apply it in the irrigation water at the crop-group stages. Never mix it with zinc, and do not neutralise it.



Measure it, dissolve it, keep it acidic

VAN gives the dose for each crop; the dealer asks VAN and does not guess. Put it into the drum at the head of the watercourse or into the drip tank, with the water flowing. It is fully soluble. Do not tank-mix it with anything that brings it towards neutral: the pH 2 to 4 keeps the phosphate in solution, and neutralised phosphate drops out before it reaches the root.

Release it slowly over the whole irrigation

Open the tap so the drum empties gradually over the whole irrigation, not all at once, and the solution reaches the wet root zone across the whole acre. Apply zinc in a separate irrigation, never in this tank: phosphate and zinc precipitate together and can block the emitter.

The stages in VAN's guidance

Fruiting vegetables 1 week after planting, then before and after flowering; potato at tuber initiation; maize at 2 to 3 leaves; orchards about 40 days after the first phosphate application. It goes in the irrigation water. Drone spraying of V-Germinator Pro is done only through VAN.

OBJECTION 1 OF 7

“4,550 for 8 litres? A bag of DAP has far more phosphate.”

What the dealer says

It does, and the bag stays in your crop plan. This is a booster dose that the irrigation water carries to the root. On calcareous soil the phosphate from the granule is locked a few centimetres from the root within weeks; the liquid goes with the water to the root at the stage the crop needs it. The pack holds 2.16 kg of P_2O_5 in solution. How much goes on an acre is VAN’s dose for your crop: I will ask VAN for it rather than guess.

The facts behind the answer

- V-Germinator Pro 8 L Rs 4,550 (VAN farmer price sheet, Aug 2026).
- 2.16 kg P_2O_5 per 8 L pack, from the declared analysis (the pack).
- No dose per acre is published; no crop plan on van.com.pk names the product.

Never say: “It replaces a bag of DAP”, “X% more available than DAP”, or a rupees-per-acre return. VAN has no trial, no published dose and no percentage.

OBJECTION 2 OF 7

“Can I skip the basal phosphate and just fertigate this?”

What the dealer says

No. It is a booster on top of the basal phosphate, never instead of it. This pack gives phosphorus at the stages VAN gives for each crop group. The season’s main phosphate comes from the basal bag applied at land preparation. A crop given only this product gets far less phosphate than it needs for the season: maize grain alone removes 6.3 kg of P_2O_5 per tonne, and the whole 8 L pack holds 2.16 kg.

The facts behind the answer

- A booster on top of the basal granule, not a substitute for it (VAN guidance, 23 July 2026).
- Maize grain removes 6.3 kg P_2O_5 per tonne (the IPNI removal tables); the 8 L pack carries 2.16 kg.
- The basal phosphate goes in at land preparation in the crop plans of field and vegetable crops.

Never say: “Fertigate this and drop the DAP.”

OBJECTION 3 OF 7

“Can I mix it with my zinc?”

What the dealer says

Never. Phosphate and zinc precipitate together as zinc phosphate: you lose both and you may block the line. This happens with every soluble phosphate; it is basic chemistry. Apply zinc in a separate irrigation or spray. Before mixing it with any other product, mix a small amount of both in a jar of water and check that nothing settles.

The facts behind the answer

- VAN’s product data: never in the same tank as zinc, or with anything that brings it towards neutral. No compatibility chart is published.
- Zinc and phosphorus interfere with each other in the plant and in the tank (VAN’s 17-nutrients page; textbook chemistry).
- Zinc phosphate is insoluble; it settles and can block an emitter.

Never say: “A little zinc is fine if you stir it.”

OBJECTION 4 OF 7

“Which stage, and how much?”

What the dealer says

VAN’s guidance gives the stages by crop group. Fruiting vegetables: 1 week after planting, then before and after flowering. Potato: at tuber initiation. Maize: at 2 to 3 leaves. Orchards: about 40 days after the first phosphate application. No crop plan gives the dose per acre yet; I will ask VAN for the dose for your crop, and you measure it. A guessed dose is the commonest error in the field.

The facts behind the answer

- Stages by crop group: fruiting vegetables 1 week after planting, then before and after flowering; potato at tuber initiation; maize at 2 to 3 leaves; orchards about 40 days after the first phosphate application (VAN guidance, 23 July 2026).
- No crop plan on van.com.pk has a V-Germinator Pro row yet.
- VAN’s product page prints the same crop-group stages (VAN guidance, 23 July 2026).

Never say or give: a dose per acre from memory, or “1 pack an acre”.

OBJECTION 5 OF 7

“Can I spray it on the leaf instead?”

What the dealer says

A dealer gives 1 route for this product: fertigation, through drip or flood irrigation water to the root. Drone spraying of V-Germinator Pro is done only through VAN. I will not give you a spray rate from a fertigation pack. If you need a foliar phosphate, ask VAN which product your crop plan names.

The facts behind the answer

- Dealers give fertigation as the route for V-Germinator Pro. Drone spraying of V-Germinator Pro is done only through VAN.
- No foliar rate or water volume for V-Germinator Pro is published.
- A foliar rate is a different figure from a fertigation dose, and neither is guessed.

Never say or give: a foliar or drone rate for V-Germinator Pro.

OBJECTION 6 OF 7

“Is it registered? What is the licence?”

What the dealer says

Yes. V-Germinator Pro is registered under Pakistan Standard PS 5469-2020 for liquid nutrient supplements, and VAN makes it under manufacturing licence CM/L-4300/2026. Every batch is tested at VAN Lab, accredited by the Pakistan National Accreditation Council (PNAC) as LAB 336, before it ships. Enter the batch number printed on the pack on van.com.pk, and you can read the certificate for that batch.

The facts behind the answer

- V-Germinator Pro: PS 5469-2020, CM/L-4300/2026 (the licence register on van.com.pk).
- Every batch is tested at VAN Lab, PNAC LAB 336, before it ships.
- A grower can read the certificate for any batch on van.com.pk by its batch number.

Never say: “The certificate reports the carrier”, or any expiry date for the licence.

OBJECTION 7 OF 7

“The crop is already purple and slow. Will this fix it this week?”

What the dealer says

Purple edges on the older leaves, thin roots, a slow crop: that is what a phosphorus-short crop looks like, and growth lost in the first weeks is not made up later. Apply it in the irrigation water at the stage VAN gives for your crop group, at the dose VAN gives for your crop. I will not promise a number of days or a yield, because VAN has measured neither.

The facts behind the answer

- A phosphorus-short crop: purple tint on the older leaves, fewer tillers, late flowering; most of the damage is done in the first weeks (VAN’s 17-nutrients page).
- The stages are given by crop group; orchards about 40 days after the first phosphate application (VAN guidance, 23 July 2026).
- No trial or response time is on VAN’s record.

Never say: “Green in a week” or any recovery figure.

ITS PLACE IN THE CROP PLANS

No published crop plan includes V-Germinator Pro yet.

So there is no rate per acre to give. VAN’s guidance gives the stages by crop group: fruiting vegetables 1 week after planting and then before and after flowering; potato at tuber initiation; maize at 2 to 3 leaves; orchards about 40 days after the first phosphate application. VAN gives the dose for each crop; the dealer asks and does not guess.

WHAT VAN DOES NOT CLAIM FOR THIS PRODUCT

A dealer who makes any of these claims is speaking for himself, not for VAN.

1. A germination, emergence or yield figure. No V-Germinator Pro trial exists.
2. That 70 to 80% of applied phosphate is fixed, or 80 to 95% of soils are deficient: neither is on VAN's record. Give the Punjab soil survey: 59.1% of samples below 7 ppm available phosphorus, the low level; 97.9% below 15 ppm, the adequate level.
3. That it replaces the basal phosphate, or works better than DAP as a phosphate source. It is a booster on top of the bag.
4. A dose per acre. None is published; the dealer asks VAN.
5. That the carrier "unlocks" the soil's phosphorus by a percentage. The carrier's sugars feed soil life; nothing is measured for this product.
6. A foliar or drone rate. Dealers give fertigation. Drone spraying of V-Germinator Pro is done only through VAN.

HOW A GROWER VERIFIES A PACK

Every batch is tested, and the batch number on the pack identifies its certificate.

1 · The batch number

Every pack carries the batch number. Read it to the grower and show him where it is printed.

2 · The certificate of analysis

On van.com.pk, a grower can read the certificate of analysis for any pack by entering the batch number printed on it. The grower does not need to post a sample.

3 · The laboratory

VAN Lab is accredited by the Pakistan National Accreditation Council (PNAC) to ISO/IEC 17025:2017, the international standard for testing laboratories, as LAB 336. Every batch is released through the laboratory before it ships. The certificate carries the laboratory's number.

What a certificate for V-Germinator Pro shows

The certificate gives the declared analysis tested for the batch: P_2O_5 against 27% w/v, with nitrogen, K_2O, magnesium, sulfur and iron where the certificate carries them. The batch number, the Pakistan Standard PS 5469-2020 and licence CM/L-4300/2026 are on the pack. The plant-based carrier has no declared figure, so the certificate does not report it. The grower reads the certificate for his own batch on van.com.pk by its batch number.

THE URDU LINES · REVIEWED 27 SEPTEMBER 2026

The lines a dealer says in Urdu.

The first line is the product's main message. Where its nutrient has one, the line gives VAN's promise word: the 1-word brand message VAN uses for that nutrient. The other 3 lines are key sentences of this deck. Each English line gives the meaning of the Urdu beside it. Send any correction to partner@van.com.pk.

Promise word REACH: "first the root, then everything".

پہلے جڑ، پھر سب کچھ

English meaning · Urdu reviewed 27 Sep 2026

It goes with the water to the root, at the stage the crop needs it.

پانی کے ساتھ جڑ تک پہنچتا ہے، جب فصل مانگے

English meaning · Urdu reviewed 27 Sep 2026

Never mix it with zinc.

زنک کے ساتھ کبھی نہ ملائیں

English meaning · Urdu reviewed 27 Sep 2026

It is on top of the basal phosphate, not in place of it.

یہ بنیادی فاسفیٹ کے اوپر ہے، اس کی جگہ نہیں

English meaning · Urdu reviewed 27 Sep 2026

TALK TO VAN

Use the address that matches your request.

For growers

kisan@van.com.pk

A crop plan, product rates, a certificate for your batch or a dealer near you.

For the trade

partner@van.com.pk

Dealers, distributors and the trade: stock, training and corrections to this deck.

For everything else

info@van.com.pk

Supply, the laboratory or a plant visit.

Helpline 0300 5003041 · www.van.com.pk

Vital Agri Nutrients (Pvt) Ltd, Lahore, has 1 helpline and 3 addresses. VAN sends no group messages; every grower who writes gets a direct reply.

V-Germinator Pro: liquid phosphate in the irrigation water at the stages for each crop group, on top of the basal bag and never with zinc.

